# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

# CHEMISTRY

Past Paper 2021 • Paper Code: 102101

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. The unit of Kc for the following reaction is, $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$.

(A) $mol.dm^{-3}$
(B) $mol^{-1}.dm^{-3}$
(C) $mol^{-1}.dm^{3}$
(D) $mol.dm^{3}$

Q2. $AlCl_3$ is an acid according to concept of.

(A) Arrhenious
(B) Lowry-Brosted
(C) Lewis
(D) All of these

Q3. Compounds which are electrons pair doner are Lewis base. Which one of the following is Lewis base,

(A) $NH_4OH$
(B) $NH_3$
(C) NaOH
(D) Nacl

Q4. Which one of the following is organic compound.

(A) $C_2H_4$
(B) CO
(C) $CO_2$
(D) $NaHCO_3$

Q5. Which one of the following is butyl group.

(A) $CH_3(CH_2)_2CH_3$
(B) $CH_3(CH_2)_2CH_2$-
(C) $CH_3(CH_2)CH_2$-
(D) $CH_3$-$CH_2$-

Q6. The general formula of alkanes is.

(A) $C_nH_{2n+2}$ (B) $C_nH_{2n-2}$

(C) $C_nH_{2n}$ (D) $C_nH_{2n+1}$

Q7. Glycogen is an example of.

(A) Fats (B) Nucleic acid

(C) Carbohydrates (D) Protien

Q8. On the basis of temperature variation the number of atmospheric regions are.

(A) Five (B) Two

(C) Three (D) Four

Q9. All salts cause the water become hard except.

(A) $Cacl_2$ (B) $Ca(HCO_3)_2$

(C) Mg $SO_4$ (D) Nacl

Q10. Matte is the mixture of.

(A) CuS and FeS (B) $Cu_2S$ and FeS

(C) $Cu_2O$ and FeS (D) $Cu_2O$ and FeO

Q11. Which gas in the following is responsible for global warming.

(A) Nitrogen (B) Carbon dioxide

(C) Oxygen (D) Ozone

Q12. How much hydrogen are there in alkane having 3 carbon atoms.

(A) Four (B) Six

(C) Eight (D) Ten

## SECTION "B"

**Time Allowed:** 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Describe the microscopic characteristics of forward and reverse reactions.

(ii) Can you distinguish between acidic and basic salts? Give example for each.

(iii) Define the functional group and give the functional group of any three organic compounds.

(iv) Write the equation for the preparation of Ethene from Ethyl alcohol and Ethyl chloride.

(v) Give any four uses of proteins..

(vi) How the acid rain affect plant and human.

(vii) Describe any two methods for the removal of permanent hardness.

(viii) Explain the roasting process for the extraction of metals.

(ix) Differenciate between the Molecular formula and structural formula with examples.

(x) Draw the dot & cross, condensed formula for pentane and Ethene.

(xi) Differenciate fats and oil.

## SECTION "C"

**Maximum Marks:** 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Derive an expression for the equilibrium constant for a general reaction.

(b) [3] Calculate the $P^{H}$ and $P^{OH}$ of 0.001 M solution of Nitric acid ($HNO_3$).

QIV.

(a) [3] How are structural formula used in organic chemistry.

(b) [4] How would you compare Troposphere with stratosphere.

QV.

(a) [4] Describe the composition of water by mass.

(b) [3] How alkyl radicals are formed. Give two examples of alkanes to explain it.

QVI.

(a) [4] What is petroleum. Name at least three fractions showing the number of carbon atoms per molecules.

(b) [3] Give the balance reactions of water with. i) Carbon ii) Chlorine iii) Silicon